

YOUR UNI: _____
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CSEE 3827 Sp'24 Midterm, **Question 0** p. 1

Midterm Solutions

CSEE W3827 - Fundamentals of Computer Systems
Spring 2024

March 5, 2024
Prof. Rubenstein

This midterm contains 16 pages: a question 0 and 3 other questions, and totals 100 points. To get full credit you must answer all questions. **NO ELECTRONIC DEVICES PERMITTED!** The time allowed is 180 minutes (3 hours).

Please answer all questions on the exam paper. Blue books are available for scrap work only. We will be scanning exams and cannot utilize any writing in the blue books.

Some advice:

- Be sure to leave some time to work on each problem. The right answer to each problem does not require a very long answer.
- Be sure to start every problem. And take some time to think about how to set the problem up before you start writing.

0. (5 pts) Do the following **on each page** of the exam:

- (a) **CLEARLY** write your **UNI**, name and P-credit TA initials at the top **of each page** of the exam in the space provided (including this page).
- (b) Answer questions on the white paper. If you need additional white pages, please raise your hand, and remember to write on the top: UNI, Name, P-Credit TA, "CSEE 3827 Sp24 Question" and indicate the question number.
- (c) label solutions (e.g., 2a, 2b, 2c or 2a, b, c) if using the additional pages.

YOUR UNI: _____

YOUR NAME: _____

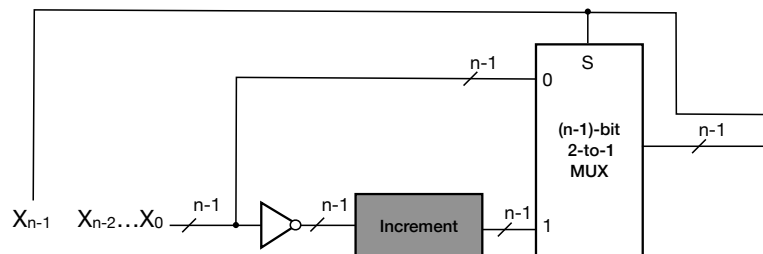
CSEE 3827 Sp'24 Midterm, **Question 1** p. 1

YOUR P-CREDIT TA (initials preferred): _____

1. (30 pts) In this problem, we will build circuits that take as input some n -bit quantity $X = X_{n-1}X_{n-2} \dots X_0$ and convert between 2's-complement and signed magnitude. In building your circuit, you may use basic gates (AND, OR, NOT, XOR), as well as basic circuits such as decoders, multiplexers, encoders, shifters, adder, incrementer, etc.

- (a) (10 pts) Draw a circuit that converts an n -bit number in 2's complement representation to its n -bit equivalent in signed magnitude.

Answer: When $X_{n-1} = 0$, no change needs to be made. When $X_{n-1} = 1$, X represents a negative value. We can find its positive equivalent by flipping all the bits and incrementing. We can then represent the result by leading with a 1 and having the remaining $n - 1$ least significant bits match the $n - 1$ least significant bits of this flipped result. Note that equivalently, we can simply start with these $n - 1$ least significant bits, flip them and add 1 to get the corresponding magnitude.



YOUR UNI: _____

YOUR NAME: _____

CSEE 3827 Sp'24 Midterm, **Question 1** p. 2

YOUR P-CREDIT TA (initials preferred): _____

- (b) (10 pts) Describe overflow circuitry for the converter from part 1a (you can draw the circuitry or just explain the values of X where overflow occurs).

Answer: Overflow only occurs when 2's complement is representing 2^{n-1} . This is the case where $X_{n-1} = 1$ and all other $X_i = 0, i < n-1$. Thus, overflow could be computed as the product (AND) of $X_{n-1}\bar{X}_{n-2}\bar{X}_{n-3}\dots\bar{X}_0$.

Note that if one assumes the incrementer outputs an overflow signal when it increments an input of all 1's, then the $(n-1)$ bit incrementer will return overflow when $X_{n-2}\dots X_0$ is all 0, i.e., the overflow can just be the AND of X_{n-1} and the overflow from the incrementer. Note this only applies to an $n-1$ bit incrementer.

YOUR UNI: _____

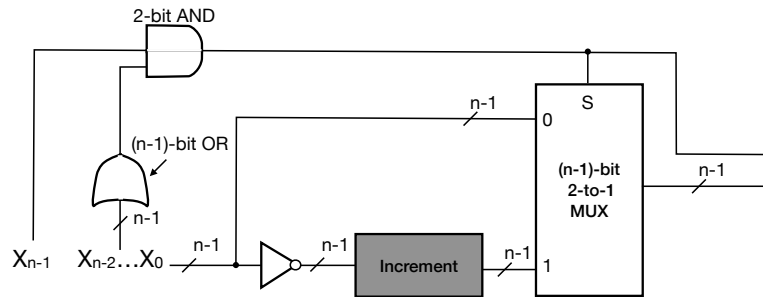
YOUR NAME: _____

CSEE 3827 Sp'24 Midterm, **Question 1** p. 3

YOUR P-CREDIT TA (initials preferred): _____

- (c) (5 pts) Draw a circuit that converts an n -bit number in signed-magnitude representation to its 2's complement representation.

Answer: Note the process is almost identical to that in part 1a i.e., if $X_{n-1} = 0$, we simply return X . If $X_{n-1} = 1$, we must flip all bits and add 1 to get the magnitude of the negative value represented in 2's complement form. The one special case is where $X_{n-1} = 1$ and all other bits are 0 (i.e., the “negative zero” in signed-magnitude). In this one case, we need to flip the sign of the high-order bit. This is accomplished in the circuit below by noting that when $X_{n-2}, X_{n-3}, \dots, X_0$ are all 0, then $X = 0$, such that we must “zero out” both the selector and the highest output bit.



YOUR UNI: _____

YOUR NAME: _____

CSEE 3827 Sp'24 Midterm, **Question 1** p. 4

YOUR P-CREDIT TA (initials preferred): _____

- (d) (5 pts) Describe the overflow circuitry for this signed-magnitude to 2's-complement converter from part 1c (as before, you can draw the circuitry or just explain the values of X where overflow occurs).

Answer: Since every value representable in signed-magnitude is also representable in 2's complement, there is no overflow.
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YOUR UNI: _____

YOUR NAME: _____

CSEE 3827 Sp'24 Midterm, **Question 2** p. 1

YOUR P-CREDIT TA (initials preferred): _____

2. (25 pts) A coin sorter examines a coin and generates four boolean signals, B, R, T and H , where the signal is TRUE if the coin is respectively **B**ig, **R**ough-edged, **T**hick, and **H**eavy. The desired output is a 2-bit quantity, XY , which is different for each type of valid coin. The table below indicates how each type of coin sets the inputs B, R, T, H , and what the corresponding output XY should be:

	Inputs				Output	
Coin	B	R	T	H	X	Y
Penny	0	0	0	0	0	0
Nickel	1	0	1	0	0	1
Dime	0	1	0	0	1	0
Quarter	1	1	1	1	1	1

- (a) (5 pts) Give algebraic expressions for X and Y in terms of inputs B, R, T, H when we only care that the outputs are correct for the four coin types listed (i.e., we “don’t care” what is output for other combinations.) (Hint: this can be done by drawing K-maps but is also easily done by eye-balling the table above. Maybe all inputs aren’t needed?)

Answer: A quick scan of the input columns shows that $X = R$ and $Y = T$ or $Y = B$.

YOUR UNI: _____

YOUR NAME: _____

CSEE 3827 Sp'24 Midterm, **Question 2** p. 2

YOUR P-CREDIT TA (initials preferred): _____

- (b) (10 pts) Suppose an additional output, V is added, which takes inputs B, R, T, H and outputs a 1 when and only when the coin inputs are **valid**, i.e., the input $BRT H$ is one of 0000, 1010, 0100, or 1111 (for all other inputs, V should equal 0). Give a simplified expression for V .

Answer:

	Inputs				Output
Coin	B	R	T	H	V
Penny	0	0	0	0	1
(none)	0	0	0	1	0
(none)	0	0	1	0	0
(none)	0	0	1	1	0
Dime	0	1	0	0	1
(none)	0	1	0	1	0
(none)	0	1	1	0	0
(none)	0	1	1	1	0
(none)	1	0	0	0	0
(none)	1	0	0	1	0
Nickel	1	0	1	0	1
(none)	1	0	1	1	0
(none)	1	1	0	0	0
(none)	1	1	0	1	0
(none)	1	1	1	0	0
Quarter	1	1	1	1	1

The corresponding k-map for V is

$$\begin{array}{c}
 \begin{array}{c} \text{H} \\ \underbrace{\hspace{1.5cm}} \end{array} \\
 \begin{array}{c} \text{B} \left\{ \begin{array}{|c|c|c|c|} \hline 1 & 0 & 0 & 0 \\ \hline 1 & 0 & 0 & 0 \\ \hline 0 & 0 & 1 & 0 \\ \hline 0 & 0 & 0 & 1 \\ \hline \end{array} \right\} \begin{array}{c} \text{T} \\ \underbrace{\hspace{1.5cm}} \end{array} \end{array} \Bigg\}^R \text{ which yields } V = \bar{B}\bar{T}\bar{H} + B\bar{R}T\bar{H} + BRT H.$$

Note the last 2 terms can be combined as $BT(R \oplus \bar{H})$ or as $BT(\bar{R} \oplus H)$.

YOUR UNI: _____

YOUR NAME: _____

CSEE 3827 Sp'24 Midterm, **Question 2** p. 3

YOUR P-CREDIT TA (initials preferred): _____

- (c) (10 pts) Suppose it is known that counterfeit coins take either the form $BRTH = 0001$ or 1100 such that the only inputs to the system would be those representing valid coins or these two invalid coins (i.e., the only inputs can be one of $BRTH = 0000, 0001, 0100, 1010, 1100$, or 1111). Give the simplified expression for V when these are the only inputs that are of concern.

Answer: Any row's output in the previous truth table that does not correspond to a valid or counterfeit coin is now a "Don't care".

	Inputs				Output
Coin	B	R	T	H	V
Penny	0	0	0	0	1
(none)	0	0	0	1	0
(none)	0	0	1	0	X
(none)	0	0	1	1	X
Dime	0	1	0	0	1
(none)	0	1	0	1	X
(none)	0	1	1	0	X
(none)	0	1	1	1	X
(none)	1	0	0	0	X
(none)	1	0	0	1	X
Nickel	1	0	1	0	1
(none)	1	0	1	1	X
(none)	1	1	0	0	0
(none)	1	1	0	1	X
(none)	1	1	1	0	X
Quarter	1	1	1	1	1

The corresponding k-map for V is

B {	H <table> <tr><td>1</td><td>0</td><td>X</td><td>X</td></tr> <tr><td>1</td><td>X</td><td>X</td><td>X</td></tr> <tr><td>0</td><td>X</td><td>1</td><td>X</td></tr> <tr><td>X</td><td>X</td><td>X</td><td>1</td></tr> </table> T				1	0	X	X	1	X	X	X	0	X	1	X	X	X	X	1	} R
	1	0	X	X																	
	1	X	X	X																	
	0	X	1	X																	
	X	X	X	1																	

Here, $V = T + \bar{H}\bar{B}$.

YOUR UNI: _____

YOUR NAME: _____

CSEE 3827 Sp'24 Midterm, **Question 3** p. 1

YOUR P-CREDIT TA (initials preferred): _____

3. (40 pts) NVIDIA is developing a new chip that's sure to make them a gazillion more dollars! To maximize profits, they want to bring down labor costs, which is where you come in!

NVIDIA has to build a processing system that can handle two types of jobs (programs): type 0 is a graphics job that takes a single clock cycle to complete, and type 1 is an AI job that takes 2 clock cycles to complete. The system consists of two processing units, a "Primary" and a "secondary" such that the system is capable of processing 2 jobs at the same time. Each clock cycle t , a job arrives to the system, with the input at time t indicating whether the job is of type 0 or type 1. At time $t + 1$ the job is given to an available processor. If the "Primary" processor is free, it accepts and processes the job to completion (i.e., using clock cycle $t + 1$ to complete the job of type 0, and clock cycles $t + 1$ and $t + 2$ to complete the job of type 1). If the "Primary" processor is busy (i.e., working on the second cycle of the type 1 job), the job is served by the "Secondary" processor.

Note that under this construction, every time a job arrives, at least one of the two processors will be available to serve it.

- (a) (15 pts) Suppose we wish to output a 1 during any clock cycle where the job being processed by the "Primary" processor has more time remaining than the job in the "Secondary" processor. A timeline example is given below:

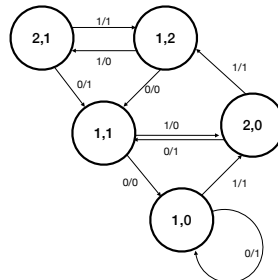
cycle #	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
input	0	0	1	0	1	1	0	0	1	0	1	1	1	0	0	1	1	0	1	0
Primary	1	1	1	2 → 1		2 → 1		1	1	2 → 1		2 → 1		2 → 1		1	2 → 1		1	2
Secondary	1	0	0	0	1	0	2 → 1		0	0	1	0	2 → 1		1	0	0	2 → 1		0
output	0	1	1	1	0	1	0	0	1	1	0	1	0	1	0	1	1	0	0	1

Note that:

- Each time t input corresponds to a job starting at time $t + 1$.
- The value shown in "Primary" and "Secondary" rows shows the remaining processing time for the job being processed. $2 \rightarrow 1$ shows a type 1 job taking 2 cycles to complete. 0 in the "Secondary" row indicates no job is being processed by the "Secondary" processor.
- Note that the "Secondary" processor will receive a type 1 job only when the "Primary" still has one cycle left to process its primary job.

Draw a state machine that can implement this process. Please make sure your state labels clearly indicate what each state represents.

Answer: In the state machine design below, state i, j represents the time remaining for the job being processed by the "primary" processor is i and the time remaining for the job processed by the secondary processor is j (with 0 representing the processor being idle). Note that aside from initial conditions, the "primary" processor will always be active, either processing the first cycle (a "2") of a type 1 job, the second cycle of a type 1 job, or the only cycle of the type 0 job. The "secondary" processor can process the first cycle (2) of a type 1 job, its second cycle, the single cycle of a type 1 job, or no job. Output is 1 whenever transitioning out of a state where $i > j$.



Note: one can in fact merge states 2,0 and 2,1 into a single state, since they have identical transitions exiting from them. However, we will accept 5-state answers as well.

YOUR UNI: _____

YOUR NAME: _____

CSEE 3827 Sp'24 Midterm, **Question 3** p. 2

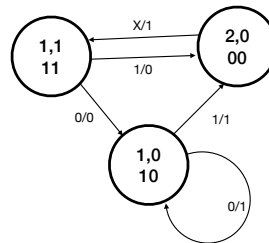
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- (b) (15 pts) Suppose we enhance the “Secondary” processor such that it can complete both job types in a single cycle, while the “Primary” processor behaves as before. The revised behavior is shown below. Note that a type 1 job being processed by the “secondary” processor completes in a single cycle, but is represented by a '2' to indicate the fact that it's a job that in the previous problem would have taken 2 clock cycles to complete:

cycle #	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
input	0	0	1	0	1	1	0	0	1	0	1	1	1	0	0	1	1	0	1	0
Primary	1	1	1	2 → 1	2 → 1	2 → 1	1	1	2 → 1	2 → 1	2 → 1	2 → 1	2 → 1	1	2 → 1	1	2	0	0	0
Secondary	1	0	0	0	1	0	2	0	0	0	1	0	2	0	1	0	0	2	0	0
output	0	1	1	1	0	1	0	1	1	1	0	1	0	1	0	1	1	0	1	1

Draw a revised state machine for this system. Hint: it can be done with fewer states than part (a). Again, please make sure your state labels clearly indicate what each state represents.

Answer: We note since the “secondary” processor always processes its job in a single cycle, we only need those states where $j = 1$ or $j = 0$. Furthermore, the system will never enter state 2, 1, since (in the previous state machine) we only entered state 2, 1 from state 1, 2 (i.e., when the secondary processor required 2 cycles to complete a job). Hence the state machine reduces to a 3-state system as drawn below. These states are also labeled with 2-bit labelings. We chose to label our states where $i = 1$ or $j = 1$ maps to 1, and otherwise maps to 0.



YOUR UNI: _____
 YOUR NAME: _____
 YOUR P-CREDIT TA (initials preferred): _____

- (c) (10 pts) Build the system for part (b) using JK flip flops. You can simply write down the boolean expressions for the output and the inputs to the JK flip-flops that you use.

The following state/excitation tables may be of assistance while designing sequential circuitry with JK Flip-flops

$J(t)$	$K(t)$	$Q(t+1)$
0	0	$Q(t)$
0	1	0
1	0	1
1	1	$\overline{Q(t)}$

$Q(t)$	$Q(t+1)$	$J_Q(t)$	$K_Q(t)$
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Answer: The table:

$A(t)$	$B(t)$	$I(t)$	$A(t+1)$	J_A	K_A	$B(t+1)$	J_B	K_B	O
0	0	0	1	1	X	1	1	X	1
0	0	1	1	1	X	1	1	X	1
0	1	0	X	X	X	X	X	X	X
0	1	1	X	X	X	X	X	X	X
1	0	0	1	X	0	0	0	X	1
1	0	1	0	X	1	0	0	X	1
1	1	0	1	X	0	0	X	1	0
1	1	1	0	X	1	0	X	1	0

Output = $\bar{B}$:

A	B I			
	00	01	11	10
0	1	1	X	X
1	1	1	0	0

$J_A = 1$:

A	B I			
	00	01	11	10
0	1	1	X	X
1	1	1	X	X

$K_A = I$:

A	B I			
	00	01	11	10
0	X	X	X	X
1	0	1	1	0

$J_B = \bar{A}$:

A	B I			
	00	01	11	10
0	1	1	X	X
1	0	0	X	X

$K_B = 1$:

A	B I			
	00	01	11	10
0	X	X	X	X
1	X	X	1	1

YOUR UNI: _____

YOUR NAME: _____

YOUR P-CREDIT TA (initials preferred): _____

CSEE 3827 Sp'24 Midterm, **Question 3** p. 4

YOUR UNI: _____

YOUR NAME: _____

CSEE 3827 Sp'24 Midterm, Extra __ for question ____

YOUR P-CREDIT TA (initials preferred): _____

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YOUR UNI: _____

YOUR NAME: _____

CSEE 3827 Sp'24 Midterm, Extra __ for question ____

YOUR P-CREDIT TA (initials preferred): _____

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YOUR UNI: _____

YOUR NAME: _____

CSEE 3827 Sp'24 Midterm, Extra __ for question ____

YOUR P-CREDIT TA (initials preferred): _____

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